Antivirus alerts: what they mean and what to do

Article KB0010084 | Owner: Security Operations | Subcategory: Antivirus / malware alert | Applies to: Windows 11 23H2/24H2 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 devices (MacBook Pro/Air) running Microsoft Defender for Endpoint

Summary

Microsoft Defender for Endpoint generates antivirus alerts when it detects suspicious files, potentially unwanted programs, or malicious activity on managed devices. This article explains what those alerts mean, distinguishes real threats from scare-ware pop-ups, provides a self-help checklist for users, and outlines the exact steps the service desk follows to investigate, contain, and remediate the issue.

Symptoms

- Defender toast notification: "Threat detected: Trojan:Win32/Emotet"
- "Windows Security" pop-up with message "Virus & threat protection found a threat"
- Browser pop-up stating "Your computer is infected" or "Call support now" (common scare-ware)
- Alert in Company Portal: "Action required: Review Defender alert"
- Protection history entry showing Quarantined, Removed, or Blocked action

Common causes

- Legitimate malware, ransomware, trojans, or potentially unwanted programs (PUAs) detected by Defender
- False positive triggered by legitimate software, scripts, or unsigned installers
- Scare-ware or tech-support scam pop-ups delivered via malicious websites
- Adware or bundled PUPs that exhibit suspicious behavior
- Behavior-based detections such as PowerShell download cradles or credential-dumping tools

Understanding the Alert

When Defender spots a file or behavior that matches a known threat signature or heuristic rule, it raises an alert, automatically quarantines the item, and shows a toast notification. The alert does not always mean the device is compromised; sometimes the detection is a false positive or a scare-ware pop-up that tries to trick the user into calling a fake support line. Recognizing the difference helps users react appropriately without unnecessary panic.

Self-Help Steps for Users

If you see a Defender alert, follow these steps before contacting the service desk. They are safe to perform on a managed laptop and will give the support team useful information.

1. **Do not interact with any pop-up** - close the window using Alt+F4 or open Task Manager and end the browser process.
2. **Open Windows Security** (click the shield icon in the system tray) and go to *Virus & threat protection* -> *Protection history*.
3. **Locate the recent alert**; note the threat name, date/time, and the action Defender took (Quarantined, Removed, Blocked).
4. **Run a quick scan**: In Windows Security -> *Virus & threat protection* -> *Quick scan*. Allow it to finish.
5. **If the alert shows "Quarantined" or "Removed" and the quick scan finds no new threats**, you can consider the issue resolved. Keep an eye out for any repeat pop-ups.
6. **If you see a scare-ware pop-up that claims infection but Defender shows no threat**, close the browser, clear browsing data (Settings -> Privacy, search, and services -> Clear browsing data), and reset the browser to default settings.
7. **When in doubt, or if the alert mentions ransomware, trojan, or a high-severity threat**, stop further troubleshooting and contact the IT Service Desk immediately via phone ext. 4357 or the Teams channel "IT Help". Leave the device powered on and connected to the network so Defender can upload logs and the team can isolate it if needed.

Service Desk Investigation Steps

L1 agents use this workflow after a ticket is created. All actions should be documented in the ticket with timestamps and

relevant IDs.

1. **Confirm device management** - verify the laptop is compliant in Intune/Company Portal.
2. **Pull Defender alert details** - sign into the Microsoft 365 Defender portal, navigate to *Alerts* -> *Queue*, filter by the device name and alert time, and capture the alert ID, threat name, and action taken.
3. **Determine legitimacy** - cross-reference the threat with threat intelligence; check if the file hash is known malware, a PUA, or a false positive.
4. **If quarantined, verify the file** - optionally submit the sample to Microsoft for deeper analysis.
5. **Run a full Defender offline scan** - either via Intune remote action *Run full scan* or instruct the user to boot into Windows Recovery and launch Microsoft Defender Offline.
6. **Evaluate scan results** - if the scan returns clean and no artifacts remain, resolve the ticket with "Threat quarantined; full scan clean."
7. **If active malware persists** - initiate containment: isolate the device through Intune, collect forensic logs (Sysinternals Autoruns, Process Explorer, network captures), and escalate to Security Operations for possible re-image.
8. **Document everything** - alert IDs, scan logs, actions taken, and any communication with the user.

Verification and Closure

Before closing the ticket, ensure the device is truly clean and the user is satisfied.

1. **Update Defender definitions** - check *Settings* -> *Windows Update* -> *Check for updates* and install any pending definition updates.
2. **Confirm no active alerts** - revisit Protection history; there should be no new entries after the scan.
3. **Ask the user to perform a quick scan** and report whether any scare-ware pop-ups reappear.
4. **Close the ticket** in the service-desk system, selecting the appropriate resolution code (e.g., "Threat quarantined - no further action" or "Device cleaned after offline scan"). Add a brief summary for the knowledge base so future analysts can reference the outcome.

When to escalate

Escalate to the Security Operations team immediately when any of the following occurs: the alert identifies ransomware, a persistent threat remains after quarantine and full scan, the device shows signs of compromise (e.g., unexpected privileged account usage, network beaconing to unknown domains, or lateral movement attempts), or the user reports a scare-ware pop-up that survives browser reset and clearing of cache. When escalating, provide the device name, user, alert timestamp, threat name, alert ID from the Defender portal, and any collected logs (Protection history export, Sysinternals output, network capture).

Keywords: Defender alert, antivirus popup, virus detected, Trojan, ransomware, scareware, tech support scam, quarantine, full scan, offline scan, Intune, Microsoft 365 Defender, Security Operations, L1, IT Help, phone extension 4357